THE CURE IS IN YOUR GENES

It's time to ditch the one-size-fits-all approach in medicine. From cancer care and cardiology to TB and rare disorders, DNA sequencing and genetic tests are helping doctors target specific pathologies and zero in on the exact treatment

BY SONALI ACHARJEE

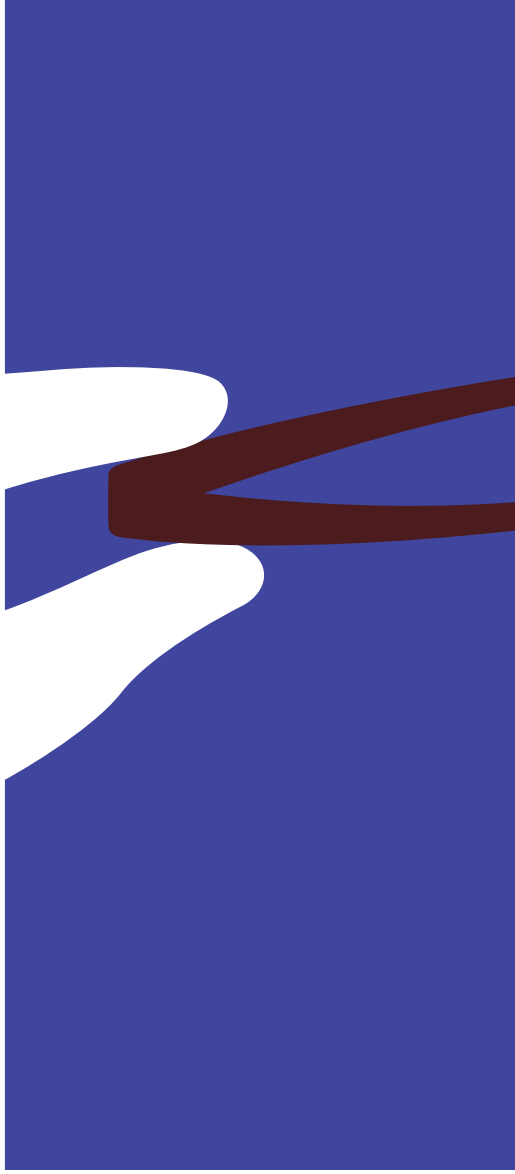


Illustration by RAJ VERMA

LAST YEAR, 11-YEAR-OLD AARAV MEHTA, FROM AHMEDABAD, stopped responding to standard treatment for a rare blood disorder. Instead of another round of broad-spectrum drugs, doctors suggested something different: whole-genome sequencing, followed by a personalised therapy designed around the exact genetic mutation causing his disease. Within weeks, Aarav's DNA was sequenced, the faulty gene pathway identified and he was put on a therapy designed for that mutation. A few months later, his symptoms stabilised. Cases like Aarav's becoming the face of a new era in Indian healthcare—one where treatment is not based on a one-size-fits-all approach developed for the average patient, but on an individual's DNA and unique disease profile.

"Instead of treating patients with the same disease in the same way, precision medicine tailors prevention and treatment based on individual differences in genes, environment and lifestyle," says Dr Vikas Maurya, HOD, Department of Pulmonology & Sleep Disorders, Fortis Hospital, Delhi. "The idea is to move from population-based medicine to personalised medicine, where treatment decisions are guided by biological data."

DNA (Deoxyribonucleic Acid) is a molecule made up of two strands of polynucleotides and holds genetic information inside cells, while genes are segments of DNA that contain instructions to build proteins, essential for our bodies to grow and function. They are the units of heredity. Genetic



"Instead of treating patients with the same disease in the same way, precision medicine tailors prevention and treatment based on individual differences in genes, environment and lifestyle"

DR VIKAS MAURYA, *HOD, Dept of Pulmonology, Fortis Hospital, Delhi*